ST. JOSEPH'S COLLEGE OF ENGINEERING

(An Autonomous Institution) • OMR, Chennai - 119

LAB RECORD

Professional English Lab Integrated

CLASS CODE: EN25C09

STUDENT DETAILS

Student Name: _____

Roll No: _____

Date: _____

Subject Name: _____

DEPARTMENT OF HUMANITIES & SCIENCES • AFFILIATED TO ANNA UNIVERSITY • CHENNAI

HARISH.M
Aspiring Software Developer | AI & Machine Learning Enthusiast
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PROFESSIONAL SUMMARY:

Enthusiastic first-year Computer Science and Engineering student aspiring to build a career in Software Development and Artificial Intelligence & Machine Learning. Possesses foundational knowledge of Python, C, and SQL, with a strong interest in problem-solving and emerging technologies. Gaining practical experience through academic projects and hackathon activities while continuously developing technical and communication skills. Motivated to learn, build real-world solutions, and grow into a skilled software professional.

TECHNICAL SKILLS:

- Programming: Python, C
- Database: SQL
- Problem Solving: Logical thinking and structured problem-solving
- Digital Tools: MS Office / Google Workspace

PRACTICAL PROJECT:

Smart Ambulance Route System | Hackathon / Academic Project

- Developed a decision-support concept for emergency ambulance routing.
- Connects emergency requests with ambulance availability, traffic conditions, route information, and hospital availability.
- Recommends a suitable route and hospital by considering both distance and emergency-service availability.
- Designed to support faster emergency decisions and help reduce delays in ambulance services.

ONLINE LEARNING:

- SWAYAM – Computing Fundamentals — Currently Pursuing (2026)

PROFESSIONAL & INTERPERSONAL STRENGTHS

- Problem Solving: Approaches problems logically and looks for practical solutions.
- Communication: Continuously developing clear and confident communication skills.
- Teamwork: Comfortable working with others toward a common project goal.

- **Adaptability: Willing to learn and adjust to new technologies and situations.**
- **Continuous Learning: Interested in improving technical knowledge and exploring emerging technologies.**

CAREER OBJECTIVE:

To develop strong skills in Software Development and AI & Machine Learning, gain practical experience through projects and continuous learning, and eventually build software solutions that solve real-world problems and make people's lives easier.

IMPRESSIVE PERSONALITY PROFILE



ANDREJ KARPATHY

Artificial Intelligence and Machine Learning Researcher & Educator

Name: Andrej Karpathy
Profession: AI/ML researcher, engineer and educator
Field: Artificial Intelligence, Machine Learning, Computer Vision and Software
Education: University of Toronto, University of British Columbia, Stanford University

1. Introduction

Andrej Karpathy is a well-known computer scientist and AI/ML expert whose work is closely connected with artificial intelligence, machine learning, computer vision and software development. His academic and professional journey shows how strong programming skills and knowledge of machine learning can be combined to build a career in advanced technology. His journey is especially inspiring for students interested in software development and AI.

2. Early Life and Education

Karpathy studied Computer Science and Physics at the University of Toronto, where he also completed a minor in Mathematics. He later pursued graduate studies in Computer Science at the University of British Columbia, focusing on machine learning, computer vision and related areas.

He continued his studies at Stanford University, where he worked toward a PhD in Computer Science with a focus on Machine Learning and Computer Vision. His academic journey demonstrates the importance of building strong fundamentals in computer science and continuously developing technical knowledge.

3. Career Journey

Karpathy gained experience through research and internships, including work with Google Research and DeepMind. His work has involved machine learning, computer vision and neural networks.

He became particularly known for making complex AI concepts easier to understand through teaching and educational content. His technical journey connects software engineering with artificial intelligence and demonstrates how programming can be used to develop intelligent systems.

4. Personality and Professional Qualities

• **Continuous Learning:** His academic and technical journey shows the importance of continuously learning new concepts and technologies.

• **Technical Curiosity:** His work across machine learning, computer vision and software demonstrates curiosity about how technology can solve difficult problems.

• **Problem Solving:** AI and machine learning require strong analytical thinking and the ability to break complex problems into smaller parts.

• **Teaching and Sharing Knowledge:** He has contributed to AI education and has helped make difficult technical concepts easier for learners to understand.

• **Innovation:** His work combines programming, mathematics and machine learning to explore new possibilities in artificial intelligence.

5. Why I Chose Andrej Karpathy

I chose Andrej Karpathy because his career is closely related to my own career interests in **Software Development and Artificial Intelligence & Machine Learning**. His journey shows how programming fundamentals can be developed into advanced skills in AI and machine learning.

As a first-year Computer Science and Engineering student, I find his career motivating because it shows that learning programming, mathematics and problem-solving step by step can lead to opportunities in advanced technology.

6. What I Learned From Him

• Build strong programming fundamentals.

• Keep learning new technologies.

• Develop problem-solving and analytical skills.

• Explore artificial intelligence and machine learning.

• Share knowledge and learn from others.

• Be curious and experiment with technology.

• Connect theoretical knowledge with practical projects.

7. Conclusion

Andrej Karpathy's journey is an inspiring example of how a strong foundation in computer science, continuous learning and curiosity can lead to a career in artificial intelligence and advanced software technology. His work motivates me to improve my programming skills and explore AI and machine learning alongside software development.

As a Computer Science and Engineering student, I want to follow a similar learning approach by **learning, building, experimenting and improving**. His journey encourages me to work towards my goal of becoming a skilled **Software Developer with expertise in AI and Machine Learning**.

References

• Stanford University – Andrej Karpathy academic and research materials
• Andrej Karpathy – Stanford academic profile and dissertation
• Andrej Karpathy – Professional and educational resources

NPTEL COURSE – DETAILED WEEK-WISE SUMMARY

Enhancing Soft Skills and Personality

Prof. T. Ravichandran | Department of Humanities and Social Sciences | IIT Kanpur

WEEK 1 – INTRODUCTION AND MINDSET

Lectures covered:

- Lecture 1: Introduction – Highlights of Developing Soft Skills Course (1–24)
- Lecture 2: Introduction – Highlights of Developing Soft Skills Course (25–48)
- Lecture 3: Mind-Set – Definitions and Types
- Lecture 4: Mind-Set – Learning Mindsets
- Lecture 5: Mind-Set – Secrets of Developing Growth Mindsets

The first week introduces the purpose and direction of the course. The opening lectures provide a preview of the earlier soft-skills course and establish why soft skills are important alongside academic or technical knowledge. The course then moves to the idea of mindset and explains that people can look at the same situation differently depending on their outlook.

The lessons distinguish different ways of thinking and introduce learning and growth-oriented mindsets. A growth mindset encourages a person to treat abilities as something that can be developed through learning, practice and experience. The week therefore creates the foundation for the rest of the course: improving personality begins with improving the way we think and respond to situations.

> **Learning outcome:** I understood the importance of having a positive learning attitude and a growth mindset. I learnt that challenges and mistakes can be treated as opportunities to learn rather than as reasons to give up.

WEEK 2 – TIME MANAGEMENT, PROCRASTINATION AND ASSERTIVENESS

Lectures covered:

- Lecture 6: Managing Time – Importance of Time and Understanding Perceptions of Time
- Lecture 7: Managing Time – Using Time Efficiently
- Lecture 8: Handling Delay – Understanding Procrastination
- Lecture 9: Handling Delay – Overcoming Procrastination
- Lecture 10: Assertiveness – Don’t Say “Yes” to Make Others Happy!

Week 2 focuses on using time effectively. The lessons examine the importance of time and the different ways people perceive it. The course encourages learners to identify how their time is actually being used and to make better decisions about important tasks. It also discusses using productive periods efficiently and avoiding unnecessary interruptions.

The next part explains procrastination, or delaying tasks that need to be completed. Instead of simply recognizing delay as a problem, the course looks at ways of overcoming it. The week then introduces assertiveness, especially the ability to communicate one's choice or refusal respectfully instead of saying yes merely to please others.

> **Learning outcome:** I learnt to plan my time, reduce unnecessary delay, identify priorities and become more assertive while communicating with others.

WEEK 3 – ASSERTIVENESS AND EMOTIONAL MANAGEMENT

Lectures covered:

- Lecture 11: Assertiveness – 2
- Lecture 12: Assertiveness – 3
- Lecture 13: Managing Negative Emotions
- Lecture 14: Channelizing Positive Emotions – Gaining Power from Positive Thinking – 1
- Lecture 15: Channelizing Positive Emotions – Gaining Power from Positive Thinking – 2

Week 3 continues the discussion on assertiveness and develops it further as a communication and personality skill. The focus is on expressing oneself appropriately rather than being overly passive or unnecessarily aggressive. Good communication requires confidence as well as consideration for other people.

The week then turns to emotional management. Negative emotions such as anger, jealousy, irritability and grief are discussed in the context of personality development. The course emphasizes that emotions are natural, but uncontrolled or excessive emotional reactions can affect judgement and behaviour. Positive thinking is introduced as a way of gaining mental strength and channelizing positive emotions.

> **Learning outcome:** I learnt to control emotional reactions, communicate more calmly and use positive thinking to handle challenges. I also understood that emotional balance is important in both personal and professional life.

WEEK 4 – PEOPLE SKILLS AND PERSONAL ATTRACTIVENESS

Lectures covered:

- Lecture 16: People Skills – What Makes Others Dislike You
- Lecture 17: People Skills – What Makes Others Like You – 1
- Lecture 18: People Skills – What Makes Others Like You – 2
- Lecture 19: People Skills – Being Attractive – 1
- Lecture 20: People Skills – Being Attractive – 2

Week 4 concentrates on interpersonal relationships and people skills. It first examines behaviours that can make a person difficult to like or work with. The lessons then move to qualities that make people more approachable and likeable, including being selfless, compassionate, kind and considerate.

The course also discusses empathy, assertiveness, optimism, encouragement, appreciating others and giving proper credit for their work. The later lectures explore attractiveness beyond physical appearance, including deeper qualities such as mindfulness, positive behaviour and meaningful connections with people. The emphasis is on developing qualities that create trust and healthy relationships.

> **Learning outcome:** I learnt that good relationships depend on empathy, respect, appreciation and considerate behaviour. I understood that personality attractiveness is strongly connected with the way we treat and understand other people.

WEEK 5 – ENGLISH SKILLS AND COMMON ERRORS

Lectures covered:

- Lecture 21: English Skills – Common Errors – 1
- Lecture 22: English Skills – Common Errors – 2
- Lecture 23: English Skills – Common Errors – 3
- Lecture 24: English Skills – Common Errors – 4
- Lecture 25: English Skills – Common Errors – 5

Week 5 is devoted to English skills and commonly made errors in everyday communication. The lectures use examples and self-check activities to help learners notice incorrect word choices, expressions and grammatical usage. Learners are encouraged to check meanings and usage rather than depending only on familiar forms.

The lessons cover commonly confused words and expressions and explain the appropriate forms through examples. The course also encourages continuous practice: making mistakes, checking them, learning the rule and trying again. This makes the module useful for improving both written and spoken English in academic and professional situations.

> **Learning outcome:** I learnt to pay more attention to commonly confused words, grammar and everyday expressions. I also understood that regular practice and correction are necessary for improving English communication.

WEEK 6 – HUMOUR, WORKPLACE COMMUNICATION AND MONEY

Lectures covered:

- Lecture 26: Significance of Humour in Communication
- Lecture 27: Humour in the Workplace
- Lecture 28: Function of Humour in the Workplace
- Lecture 29: Money and Personality
- Lecture 30: Managing Money

Week 6 examines the role of humour in communication. The course explains that appropriate humour can make communication more pleasant, create a lighter atmosphere and help people connect. At the same time, humour needs sensitivity: workplace communication should avoid jokes that insult, belittle or offend others.

The workplace lectures discuss how humour can sometimes support cooperation, reduce tension and help people see situations from another perspective. The course also connects personality with attitudes toward money and introduces responsible money management. The emphasis is on developing balanced behaviour rather than allowing money to control personal decisions.

> **Learning outcome:** I learnt to use humour carefully and appropriately in communication. I also gained an understanding that responsible financial habits and balanced attitudes toward money are part of personal development.

WEEK 7 – HEALTH, EXERCISE, DIET, SLEEP AND RELATIONSHIPS

Lectures covered:

- Lecture 31: Health and Personality
- Lecture 32: Managing Health – Importance of Exercise
- Lecture 33: Managing Health – Diet and Sleep
- Lecture 34: Love and Personality
- Lecture 35: Managing Love

Week 7 connects physical and emotional well-being with personality. The course discusses how health can influence a person's energy, behaviour and overall quality of life. Exercise is presented as an important part of maintaining health, while diet and sleep are discussed as everyday habits that support well-being.

The final part of the week considers love and personality and the importance of managing emotions and relationships responsibly. The overall message is that personality development is not only about communication at college or work; it also includes taking care of one's body, mind and personal relationships.

> **Learning outcome:** I learnt that healthy habits such as regular exercise, suitable food habits and adequate sleep can support better functioning. I also understood the importance of emotional balance and responsible behaviour in relationships.

WEEK 8 – ETHICS, ETIQUETTE, MEMORY AND ENVIRONMENT

Lectures covered:

- Lecture 36: Ethics and Etiquette
- Lecture 37: Business Etiquette
- Lecture 38: Managing Mind and Memory
- Lecture 39: Improving Memory
- Lecture 40: Care for Environment

The final week focuses on responsible personal and professional behaviour. Ethics and etiquette introduce the importance of behaving appropriately, respectfully and responsibly. Business etiquette extends these ideas to professional environments, where communication, conduct and consideration for others influence how a person is perceived.

The week also discusses managing the mind and memory and presents ways of improving memory. The final lecture broadens personality development to include responsibility toward the environment. Thus, the course ends by connecting individual development with professional conduct, mental discipline and social responsibility.

> **Learning outcome:** I learnt the importance of ethical behaviour and professional etiquette. I also understood that improving memory requires attention and practice, and that responsible personality development includes caring for the environment.

OVERALL LEARNING FROM THE COURSE

After going through the eight weeks, I understood that becoming a successful professional requires more than technical knowledge. The course connects mindset, time management, communication, emotional control, people skills, language ability, health, ethics and responsible behaviour. These skills are directly useful for a Computer Science student because they can improve teamwork, project communication, presentations, academic discipline and future workplace performance.

The course also encouraged continuous self-improvement. I learnt that communication and personality are not fixed qualities; they can be strengthened through awareness, practice, feedback and consistent effort. I can apply these lessons while working on academic projects, participating in team activities and preparing for my future career in software development and technology.

Source: NPTEL, Enhancing Soft Skills and Personality, Prof. T. Ravichandran, Department of Humanities and Social Sciences, IIT Kanpur.

SWOC ANALYSIS

Building My Career as a Software Developer

Candidate: Harish.M

B.E. Computer Science and Engineering • 1st Year

Career Goal: Software Developer / Software Engineer

"Using technology to solve real-world problems and create software that makes people's lives easier."

SWOC ANALYSIS

Career Vision & Objective

VISION

Grow from a first-year CSE student into a skilled and confident Software Developer, capable of building useful software solutions for real-world problems.

OBJECTIVE

Strengthen programming fundamentals, improve problem-solving, gain practical project experience, and build a strong foundation for a successful career in software development.

CAREER DIRECTION

Learn → Build → Practice → Improve → Become a Software Developer

SWOC ANALYSIS

SWOC Matrix at a Glance

STRENGTHS

- Programming foundation in Python and C
- Problem-solving interest
- Willingness to learn
- Project experience

WEAKNESSES

- Limited industry experience
- Developing advanced skills
- Need for better consistency
- Public speaking confidence

OPPORTUNITIES

- Software career demand
- AI & emerging technologies
- Hackathons and projects
- GitHub & LinkedIn

CHALLENGES

- Rapid technology change
- Strong competition
- Balancing academics & skills
- Keeping up with AI

Harish.M • B.E. CSE

SWOC ANALYSIS

My Strengths

PROGRAMMING FOUNDATION

Foundation in Python and C.

PROBLEM-SOLVING INTEREST

Enjoy understanding problems and finding logical solutions.

WILLINGNESS TO LEARN

Interested in learning new technologies and improving continuously.

PRACTICAL PROJECT EXPERIENCE

Started applying knowledge through projects such as the Smart Ambulance Route System.

PROFESSIONAL PROFILE

Developing GitHub and LinkedIn to document learning and showcase work.

SWOC ANALYSIS

My Weaknesses

LIMITED INDUSTRY EXPERIENCE

As a first-year student, I have not yet had industry or internship experience.

ADVANCED SKILLS IN PROGRESS

Still developing advanced programming, tools and technology knowledge.

CONSISTENCY

Need to improve consistency in coding practice and learning goals.

PUBLIC SPEAKING

Working on becoming more confident and effective while speaking.

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SWOC ANALYSIS

My Opportunities

GROWING SOFTWARE DEMAND

Technology growth creates opportunities for a software career.

AI & EMERGING TECHNOLOGIES

Learning AI and modern technologies can expand my technical knowledge.

PROJECTS & HACKATHONS

Apply learning, solve real-world problems and gain practical experience.

GITHUB & LINKEDIN

Showcase projects, skills, learning progress and achievements.

ONLINE LEARNING

Courses and technical communities support learning beyond the classroom.

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SWOC ANALYSIS

My Challenges

RAPIDLY CHANGING TECHNOLOGY

Continuously learn and update my skills as tools evolve.

STRONG COMPETITION

Build strong technical skills and a good project portfolio.

BALANCING PRIORITIES

Manage academics with coding, projects and other activities.

PRACTICAL EXPERIENCE

Gaining real-world exposure is still a challenge as a first-year student.

KEEPING UP WITH AI

Use AI effectively while maintaining strong programming fundamentals.

SWOC ANALYSIS

Action Plan: Improving My Weaknesses

LIMITED EXPERIENCE

Work on projects, participate in hackathons, and look for internships as I progress.

ADVANCED SKILLS

Strengthen fundamentals first, then gradually learn advanced technologies and tools.

CONSISTENCY

Set regular time for coding practice and technical learning.

PUBLIC SPEAKING

Practise presentations, discussions and speaking activities to build confidence.

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SWOC ANALYSIS

Action Plan: Using My Opportunities

BUILD PROJECTS

Create practical projects that solve real-world problems and add them to GitHub.

JOIN HACKATHONS

Improve teamwork, creativity, problem-solving and practical development skills.

LEARN EMERGING TECH

Gradually explore AI, databases, cloud computing and relevant technologies.

STRENGTHEN PROFILE

Keep LinkedIn and GitHub updated with projects, skills and achievements.

LEARN BEYOND CLASS

Use NPTEL, online courses and technical communities for continuous learning.

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SWOC ANALYSIS

My 4-Year Career Milestones

Roadmap to becoming a Software Developer

2026–27

YEAR 1

BUILD FOUNDATIONS

Python & C fundamentals

Problem-solving

Small projects

2027–28

YEAR 2

EXPAND KNOWLEDGE

Data structures & algorithms

Databases

Hackathons & team projects

2028–29

YEAR 3

GAIN EXPERIENCE

Advanced projects

Internship preparation

Strengthen GitHub portfolio

2029–30

YEAR 4

START MY CAREER

Final-year project

Placement preparation

Technical interviews

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Conclusion & Key Takeaways

KNOW MYSELF

SWOC helps me understand my strengths and identify areas to improve.

LEARN & BUILD

Strengthen programming fundamentals while gaining practical experience.

USE OPPORTUNITIES

Make use of GitHub, LinkedIn, online learning and emerging technologies.

KEEP IMPROVING

Continuous learning, practice and adaptability are essential for my career.

LEARN → BUILD → IMPROVE → GROW → BECOME A SOFTWARE DEVELOPER

SPEAK & IMPROVE – LEARNING REFLECTION

Harish.M | B.E. Computer Science and Engineering



My Result: CEFR Level B1

What I Learnt

- I learnt how to express my ideas clearly and confidently in English.
- I improved my ability to organize my thoughts before speaking.
- I learnt that pronunciation, fluency, and clear communication are important for effective speaking.
- The test helped me understand my current speaking level and identify areas for improvement.

What I Gained

- I gained more confidence in speaking English and expressing my opinions.
- I gained a better understanding of my strengths and areas that need practice.
- I gained motivation to practise speaking regularly and improve my communication skills.
- The B1 result gave me a clear starting point for further improvement.

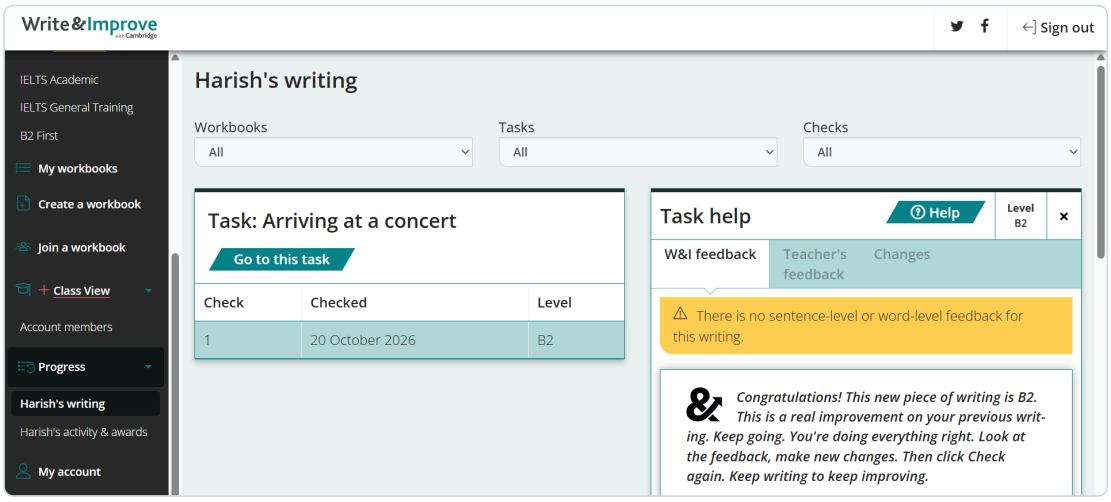
My Improvement Plan

- Practise speaking English every day for a few minutes.
- Listen to English conversations and learn useful expressions.
- Record my answers and review my pronunciation and fluency.
- Continue using Speak & Improve to track my progress.

Reflection: This activity helped me understand where I am now and motivated me to become a more confident English speaker.

WRITE & IMPROVE — LEARNING REFLECTION

Harish M | B.E. Computer Science and Engineering



Task: Arriving at a concert

Write & Improve with Cambridge Evaluation

CEFR B2

Check	Date Checked	Evaluated Level
Check 1	20 October 2026	**B2 (Vantage / Upper-Intermediate)**

Feedback & Encouragement:
"Congratulations! This new piece of writing is B2. This is a real improvement on your previous writing. Keep going. You're doing everything right. Look at the feedback, make new changes. Then click Check again. Keep writing to keep improving."

What I Learnt

- I learnt how to structure descriptive writing clearly and effectively in English.
- I enhanced my ability to select appropriate vocabulary suited for personal reflections and descriptions.
- I learnt how automated writing evaluation helps identify grammatical correctness and sentence flow.
- The assessment helped me understand my writing capability at the B2 CEFR level.

What I Gained

- I gained greater confidence in expressing thoughts systematically through written English.
- I gained a clearer awareness of my writing strengths and structural accuracy.
- I gained motivation to write more complex sentences and refine my stylistic choices.
- Achieving the B2 benchmark provided me with a solid foundation for target academic and professional writing.

My Improvement Plan

- Write short daily essays or paragraph reflections to maintain consistency.
- Review teacher and automated feedback carefully to eliminate repetitive errors.
- Expand my vocabulary by incorporating B2 and C1 level linking words and transition phrases.
- Continue using Write & Improve to track incremental progress across multiple topics.

> ***Reflection:*** *This activity provided meaningful feedback on my written communication skills and motivated me to strive for higher accuracy and fluency in academic and professional writing.*

EN25C09: PROFESSIONAL ENGLISH – I

Model Exam I · Practice Worksheet

PART A – 15 × 1 = 15 Marks

1. Simple Present Tense
Answer: monitors

2. Simple Past Tense
Answer: recorded

3. Comparative Adjective
Answer: stronger

4. Modal of Obligation
Answer: must

5. Modal of Capability
Answer: can

6. Relative Clause
Answer: The laboratory developed a new testing device that measures material strength with high precision.

7. First Conditional
Answer: rises

8. Third Conditional
Answer: would not have failed

9. Infinitive of Purpose
Answer: The mechanics installed dual damper springs to isolate chassis resonance.

10. Gerund after a Preposition
Answer: optimizing

11. Impersonal Passive Voice
Answer: It is reported that 50 mL of hydrochloric acid was poured into the titration flask.

12. Adverb of Manner
Answer: smoothly

13. Reported Speech
Answer: The project supervisor asked the engineer whether they had calibrated the pressure sensor before testing.

14. Countable and Uncountable Nouns
Answer: The laboratory purchased several new pieces of equipment for conducting material testing.

15. Irregular Comparative

Answer: worse

PART B – 10 × 2 = 20 Marks

1. Simple Past Tense

Answer: The engineer calibrated the pressure sensor and verified its accuracy.

2. Comparative Sentences

Answer: Portability: A laptop is more portable than a desktop computer because it is lightweight and easy to carry.
Screen size: A desktop computer usually has a larger screen than a laptop, which makes it more comfortable for long working sessions.

3. Relative Clause

Answer: The engineer designed a new cooling system that reduces the temperature of industrial machines.

4. Conditional Sentences

Answer: (a) If the battery voltage drops below the required level, the system will give a warning and stop the device safely.
(b) If the technician had checked the connections earlier, the equipment would not have stopped working.

5. Impersonal Passive Voice

Answer: It was reported that the solution had been heated to 80°C before testing.

6. Reported Speech

Answer: The project manager asked the technician whether they had inspected the pressure valves before starting the compressor.

7. Uncountable Nouns

Answer: (a) The engineer received a lot of feedback on the new design.
(b) The students received several useful pieces of advice from their teacher.

8. Infinitive of Purpose / Gerund

Answer: (a) The technician wears safety gloves to protect their hands.
(b) The cooling fan operates continuously to prevent the motor from overheating.

9. Adverbs of Manner

Answer: (a) The machine operates efficiently to reduce energy consumption.
(b) The technician checked the electrical connections carefully before starting the machine.

10. Modal Auxiliaries

Answer: (a) The robotic arm can lift heavy components with precision.
(b) Visitors must not enter the testing area without authorization.

PART C – 5 × 13 = 65 Marks

For each question with an either/or choice, one option is answered in full. The selected answers are: 26(a), 27(a), 28(b), 29(a) and 30(a).

Question 26(a): Eight imperative safety instructions for safe internet use

1. Use strong, unique passwords for your email, banking and social media accounts.

2. Turn on two-factor authentication wherever it is available.
3. Never click suspicious links or download attachments from unknown senders.
4. Check the website address before entering passwords or personal information.
5. Do not share your passwords, one-time passwords or private details with anyone.
6. Keep your phone, computer, browser and antivirus software updated.
7. Use secure Wi-Fi for important activities, and avoid online banking on public networks.
8. Report suspicious messages, fake profiles and cyberbullying instead of responding to them.

Question 27(a): A traditional art form in my region and how modern initiatives can preserve it

Kolam is a traditional art form that can be seen in many homes across Tamil Nadu. It is usually drawn in front of the house, especially during festivals and special occasions. Although it may look simple, kolam reflects creativity, patience and the cultural values of the people who practise it.

Traditionally, women draw kolams early in the morning after cleaning the entrance. They use rice flour or white kolam powder to create patterns with dots, lines and curves. During Pongal and other celebrations, colourful designs and large rangolis make the streets and homes look bright. In the past, the skill was often learned by watching elders and practising every day. Kolam is also connected with welcoming guests and beginning the day with a positive feeling.

Today, this art form can be preserved by introducing it to children through school activities, cultural programmes and competitions. Local organisations can conduct workshops where experienced artists teach traditional patterns. Social media and short tutorial videos can help young people learn different designs and understand their meaning. Colleges can also include kolam demonstrations during cultural festivals. At the same time, people should continue using traditional materials such as rice flour whenever possible, rather than depending only on plastic stencils and artificial decorations.

Kolam is more than a decoration at the entrance of a house. It carries memories, family traditions and a sense of belonging. By practising it and teaching it to younger generations, we can keep this beautiful part of Tamil culture alive.

Question 28(b): Formal letter requesting permission to attend an AI and Emerging Technologies workshop

From:

Harish M.

B.E. CSE – C

St. Joseph’s Institution of Technology

4 November 2026

To:

The Principal

St. Joseph’s Institution of Technology

Subject: Request for permission to attend a workshop on Artificial Intelligence and Emerging Technologies

Respected Sir/Madam,

I am a first-year student of B.E. Computer Science and Engineering. I am writing to request permission to participate in a technical workshop on Artificial Intelligence and Emerging Technologies organised by a reputed institution.

The workshop will help me understand the basic ideas behind artificial intelligence, machine learning and other emerging technologies. I hope to learn how these technologies are used to solve real-world problems. The sessions and practical activities will also give me an opportunity to improve my technical knowledge, interact with students from other institutions and learn from experienced professionals.

Taking part in this programme will support my academic learning and encourage me to explore new areas in computer science. I will make sure that I collect the workshop details, complete any required registration and follow the instructions given by the organisers. I also request permission to attend the programme on the scheduled date. If necessary, I will submit the invitation and event schedule to the department in advance.

I assure you that I will maintain discipline and share the important points I learn with my classmates after the workshop. I kindly request you to consider my request and grant me permission to participate.

Thank you.

Yours faithfully,

Harish M.

B.E. CSE – C

Question 29(a): AI-based assessment versus traditional teacher-based assessment in engineering education

Assessment is an important part of engineering education because it helps students and teachers understand how well a topic has been learned. Today, colleges can use artificial intelligence (AI) tools to assess some tasks, while traditional assessment depends mainly on teachers. Both methods have advantages and limitations.

AI-based assessment can check objective answers, quizzes and some programming tasks very quickly. It can give students immediate feedback and help them identify mistakes without waiting for the teacher to correct every answer. It can also apply the same marking rules to many students, which may improve consistency. However, an AI system may misunderstand an unusual answer or fail to recognise a correct solution written in a different way. Its accuracy also depends on the quality of the questions, data and software.

Teacher-based assessment takes more time, especially when a large number of answer sheets must be checked. Even so, teachers can understand the reasoning behind a student's answer. In engineering subjects, they can give credit for a good method, practical effort or a creative solution even when the final answer is not perfect. Teachers can also consider a student's explanation, teamwork and laboratory performance. However, personal bias or differences in marking may sometimes affect fairness.

AI can make assessment faster and more consistent, while teachers provide human understanding and judgement. The best approach is to combine both methods. AI can support quick checking and progress tracking, but teachers should review important answers and make the final judgement. This balanced method can make engineering assessment more useful, fair and supportive of student learning.

Question 30(a): Personal email to a school friend about the first semester of engineering

To: schoolfriend@example.com

Subject: How my first semester of engineering is going

Hey!

How are you? I hope everything is going well with you. I wanted to tell you about my first semester as an engineering student. College life has been quite different from school, and I am slowly getting used to the new routine.

I have made a few good friends in my class. We attend lectures together, discuss difficult topics and help each other with assignments. The classroom activities are interesting, especially when our teachers connect the lessons with real-life examples. The laboratory sessions are also useful because we get to try things instead of only reading about them. I have started learning programming and practising basic problems, which is challenging but enjoyable.

At first, adjusting to college was not very easy. The timetable is busy, and we have to manage assignments, lab records and tests on our own. I am learning to plan my time better and complete my work before the deadlines. I also enjoy the freedom and responsibility that come with college life.

Our semester examinations are coming up, so I have started revising the subjects regularly. I am preparing short notes, practising important questions and clearing my doubts with classmates and teachers. I hope to do well in the exams and begin the next semester with more confidence.

That is all about my college experience so far. Tell me how things are going at your end. It would be great to meet during the holidays!

Take care,

Harish

PROFESSIONAL ENGLISH – I

Model I Practice Worksheet

PART A – GRAMMAR EXERCISES

Exercise 1: Simple Present Tense (Subject–Verb Concord)

1. recognize
2. requires
3. transmit
4. agrees
5. form

Exercise 2: Simple Past Tense

1. conducted
2. deployed
3. opened
4. published
5. detected

Exercise 3: Comparative Adjectives

1. higher
2. safer
3. faster
4. more durable
5. lower

Exercise 4: Modals of Obligation and Necessity

1. must
2. must
3. have to
4. must
5. has to

Exercise 5: Modals of Capability and Possibility

1. can
2. might
3. can
4. may
5. can

Exercise 6: Relative Clauses

1. The civil engineer who designed the suspension cable bridge received the national infrastructure award.

2. The spectrometer, which analyzes infrared emission spectra, is mounted inside the cleanroom vacuum chamber.

3. The technical specialist whose patent on pneumatic grippers was approved was reassigned to the robotics lab.

4. This microcontroller, which features ultra-low-power sleep states, is ideal for battery-operated remote sensors.

5. The field researchers who surveyed the wind turbine farm submitted their acoustic emission assessment yesterday.

Exercise 7: If-Clauses (Type 1)

1. exceeds

2. fails

3. engages

4. rise

5. registers

Exercise 8: If-Clauses (Type 3)

1. had started

2. had lubricated

3. would have been prevented

4. would have tripped

5. had remained

Exercise 9: Infinitives of Purpose

1. The mechanics installed dual damper springs to isolate chassis resonance.

2. Technicians purge the fuel line with dry nitrogen gas to remove residual flammable vapours.

3. We introduced an optical encoder to determine spindle velocity accurately.

4. The developers wrote an automated load-testing script to simulate ten thousand concurrent connections.

5. Gold plating is applied to terminal connectors to prevent atmospheric corrosion.

Exercise 10: Gerunds after Prepositions

1. redesigning

2. replacing

3. reducing

4. cleaning

5. filtering

Exercise 11: Impersonal Passive Voice

1. It is reported that 50 mL of hydrochloric acid was poured into the titration flask.

2. It is reported that the robotic arm was tested under a 50 kg dynamic payload.

3. It was reported that the optical pyrometer had been calibrated before ignition.

4. It was observed that severe cavitation damage had occurred along the impeller blades.

5. It is stated that volatile solvents are stored in flame-resistant containment cabinets.

Exercise 12: Adverbs of Manner

1. precisely
2. periodically
3. rapidly
4. firmly
5. smoothly

Exercise 13: Reported Speech (Questions)

1. The quality manager asked the technician whether the insulation resistance test had been documented.
2. The lead architect enquired whether the server cluster could handle double the transaction throughput.
3. The field auditor asked the safety engineer whether they had inspected the gas leakage relief lines.
4. The client asked the team whether the firmware update would fix the intermittent battery drain.
5. The instructor asked the lab group why the multimeter had shown fluctuating voltage readings.

Exercise 14: Countable and Uncountable Nouns

1. The procurement officer purchased several new machines and pieces of equipment for the workshop.
2. The laboratory generated a huge amount of test data during the wind tunnel trials.
3. We need three more software licences to complete the finite element analysis on the model.
4. The field report provided valuable information regarding the soil composition.
5. All chemical apparatus and glassware must be washed thoroughly after the experiment.

Exercise 15: Irregular Comparative Adjectives

1. worse
2. better
3. farther
4. less
5. further

PART B – GRAMMAR AND TECHNICAL COMMUNICATION

Exercise 1: Simple Past Tense Conversion

1. The technician inspected the solar arrays and recorded the peak power conversion value.
2. The chemical analyzer detected trace mercury levels and sent an automated alert to the environmental dashboard.
3. Maintenance crews lubricated the heavy conveyor bearings and replaced the worn rubber belts before the morning shift.
4. The supervisory software polled field sensors every five milliseconds and archived temperature trends in a secure database.
5. The testing team applied a 500-volt dielectric pulse across the stator winding and monitored it for insulation breakdown.

Exercise 2: Comparative Adjectives – Technical Metrics

1. Cloud storage usually has a lower initial setup cost than on-premise physical servers. On-premise servers, however, may offer more direct control over the equipment.
2. Cloud storage is more scalable than on-premise servers because storage capacity can be increased when needed. Physical servers may require additional hardware to expand.

3. Fibre-optic cables provide higher data-transfer bandwidth than copper twisted-pair cables. Copper cables are generally more limited when very large amounts of data must be transmitted.

4. Fibre-optic cables are less affected by electromagnetic interference than copper cables. This makes fibre a better choice in environments with electrical noise.

5. Electric vehicles produce fewer tailpipe emissions than conventional ICE vehicles. In fact, EVs have no tailpipe emissions while driving.

6. Electric vehicles often have a higher initial purchase price than comparable petrol vehicles. However, their running and maintenance costs may be lower.

7. Hydraulic actuators can produce greater force than many electromechanical actuators of a similar size. Electromechanical actuators are usually cleaner because they do not use hydraulic fluid.

8. Hydraulic actuators have a higher risk of fluid leakage. Electromechanical actuators are less likely to create this type of mess.

9. Relational databases usually have a more rigid schema than document databases. This makes SQL databases suitable for data that follows a fixed structure.

10. Document databases are often more flexible and may offer better horizontal scaling for certain workloads. The best choice depends on the application's query and consistency needs.

Exercise 3: Relative Clauses

1. The algorithm, which was patented by the aeronautics team, analyzes microscopic cracks in turbine blades.

2. The lead chemical engineer, who noticed an abnormal pressure spike in chamber two, authorized the emergency boiler shutdown.

3. The laboratory where high-voltage insulator puncture tests are conducted houses a 400 kV impulse voltage generator.

4. We engaged an external cybersecurity consultant whose firmware vulnerability audit exposed three backdoor ports.

5. The supervisory system uses an artificial neural network that predicts transformer insulation life based on dissolved gas concentrations.

Exercise 4: If-Clauses – Technical Conditionals

1. (a) If the operating temperature exceeds 85°C, the cooling system will start automatically. (b) If the operator had pressed the emergency shutdown switch immediately, the equipment would have stopped before further damage occurred.

2. (a) If the supply voltage drops below the 200V threshold, the control unit will disconnect the load. (b) If the network administrator had enabled multi-factor authentication, the unauthorized login might have been prevented.

3. (a) If toxic carbon monoxide concentrations surpass 50 ppm in the tunnel, the warning alarm will sound and the ventilation system will start. (b) If the mechanical team had checked the bearing lubrication schedule, the bearing failure might have been avoided.

4. (a) If the AGV detects an obstacle within 1 meter, it will slow down and stop. (b) If the thermal sensor had registered the runaway reaction in time, the safety system would have activated earlier.

5. (a) If the software receives an unauthenticated API token, it will reject the request. (b) If the civil inspectors had reviewed the core sample density tests, they would have identified the weak section earlier.

Exercise 5: Impersonal Passive Voice

1. It is reported that the crude extract was distilled at a constant temperature of 60°C.

2. It was observed that the cross-section of the weld seam had been polished with fine diamond paste.

3. It is reported that fifteen cast-iron pump casings were rejected because of internal porosities.

4. It was recorded that the resistance across the bridge circuit had been measured using a calibrated digital ohmmeter.

5. It is reported that the emergency aerodynamic braking flaps were activated at an altitude of 3,000 meters.

Exercise 6: Reported Speech

1. The safety supervisor asked the intern whether they had verified the grounding connections on the generator.

2. The plant director asked the site engineer whether the emergency effluent treatment plant had received clearance.

3. The IT security lead asked the developer why they had disabled the firewall rule on the database server.

4. The client asked the project manager whether the bridge structure could withstand Category 4 hurricane winds.

5. The research guide asked the postgraduate scholar how many sample trials they had conducted under cryogenic conditions.

Exercise 7: Countable and Uncountable Nouns

1. (a) We need to install several new software packages and machines in the CAD laboratory. (b) The team collected plenty of useful information from the field test.

2. (a) The technician ordered ten new pieces of apparatus and modern electronic equipment for the antenna lab. (b) High-speed train operations require advanced transport infrastructure and a reliable electrical power supply.

3. (a) The environmental sensor gathers extensive atmospheric data and pollution readings every minute. (b) The construction crew unloaded tons of sand, cement and steel debris at the worksite.

4. (a) All electrical hardware and wiring must comply with local safety legislation. (b) The data scientist gained useful knowledge after running the machine-learning clustering process.

5. (a) Laboratory staff must clear away all glassware and acidic waste before leaving the premises. (b) The project failed due to a lack of proper funding and technical research.

Exercise 8: Infinitives of Purpose and Gerunds

1. (a) A pressure relief valve is installed to vent excess boiler steam. (b) The automated script runs every midnight for archiving legacy server logs.

2. (a) Automotive engineers installed an intercooler to reduce the compressed-air intake temperature. (b) The technicians analyzed the structural stress points by placing piezoelectric strain gauges along the girder.

3. (a) High-performance heatsinks feature copper heatpipes to conduct heat away from the silicon die. (b) The factory cut water consumption by 30% by implementing a closed-loop recycling filtration system.

4. (a) Safety guards are mounted over rotating flywheel belts to prevent accidental worker contact. (b) The algorithm optimizes memory bandwidth by pre-fetching sequential array indices into L2 cache.

5. (a) The electronics lab installed a UPS to protect servers against sudden voltage drops. (b) The metallurgy group enhanced surface hardness by diffusing nitrogen atoms into the steel casing.

Exercise 9: Adverbs of Manner

1. (a) automatically (b) flawlessly

2. (a) accurately (b) instantly

3. (a) continuously (b) cautiously

4. (a) consistently (b) catastrophically

5. (a) periodically (b) smoothly

Exercise 10: Modals

1. (a) can (b) must not

2. (a) must (b) might
3. (a) must not (b) could
4. (a) should (b) must
5. (a) can (b) must not

TOPIC 1 – IMPERATIVE SAFETY INSTRUCTIONS

Question 1: Chemistry and Materials Testing Laboratory

1. Wear a lab coat, safety goggles and suitable gloves before starting any experiment.
2. Read the chemical label and safety instructions before using a reagent.
3. Keep food and drinks outside the laboratory.
4. Use the fume hood when working with chemicals that may release harmful vapours.
5. Never touch chemicals directly or smell them closely.
6. Keep chemical containers closed and return them to their correct storage area.
7. Clean spills only by following the laboratory's approved procedure and inform the supervisor immediately.
8. Switch off equipment after use and wash your hands before leaving the laboratory.

Question 2: High-Voltage Electrical Machines and Drives Laboratory

1. Enter the laboratory only with the instructor's permission.
2. Wear the required protective equipment and keep the work area dry.
3. Check cables, insulation and connections before switching on the supply.
4. Keep your hands away from exposed electrical terminals.
5. Never change a circuit connection while the power supply is on.
6. Use the emergency stop only when required and report any fault immediately.
7. Keep bags, water bottles and other unnecessary objects away from the test area.
8. Switch off and isolate the power supply before inspection or maintenance.

Question 3: CNC Machining Workshop

1. Wear safety goggles, safety shoes and other required protective equipment.
2. Tie back long hair and keep loose clothing and jewellery away from moving machinery.
3. Check that the workpiece and cutting tool are properly secured before operation.
4. Keep machine guards in place while the spindle is running.
5. Never touch hot metal chips with bare hands; use the approved tools to clear them after the machine stops.
6. Keep the floor clean and report coolant or oil spills immediately.
7. Do not leave a running machine unattended.
8. Stop the machine and inform the instructor if you notice unusual noise, vibration or damage.

TOPIC 2 – FREE WRITING: FESTIVALS AND CULTURAL CELEBRATIONS

Question 1: A Traditional Harvest Festival in My Region

Pongal is one of the most loved harvest festivals in Tamil Nadu. It is celebrated in January to thank the Sun, nature, farmers and cattle for supporting agriculture. The festival brings families and neighbours together and reminds us of

the importance of farming. Before the celebration, people clean their homes and decorate the entrances with colourful kolams. On the first day, families prepare sweet Pongal using newly harvested rice, milk and jaggery. The dish is traditionally cooked in a decorated pot, and everyone waits for it to boil over as a sign of prosperity. Sugarcane, banana and other seasonal foods are also shared. In many villages, people decorate their cattle and thank them for their role in farming. Children enjoy traditional games, music and visits to relatives. Today, we can celebrate Pongal in a more eco-friendly way without losing its meaning. We can use clay or reusable plates instead of disposable plastic, avoid plastic decorations and choose natural materials for kolams. Food should be prepared according to the number of guests so that it is not wasted. Organic waste can be composted, and reusable decorations can be stored for the next year. We can also support local farmers by buying their products. Pongal is not only a festival; it is a reminder to respect nature and the people who grow our food. Sustainable habits help us protect that message for future generations.

Question 2: A Traditional Temple Car Festival

A traditional temple car festival is a special event in many towns across Tamil Nadu. On the festival day, the temple streets become lively with music, decorations and crowds of devotees. The beautifully decorated temple car is taken through the streets, and people gather to take part in the procession. Families offer flowers, fruits and prayers, while volunteers help guide the crowd. Traditional food such as lemon rice, curd rice, sundal and sweet Pongal is often shared with visitors. These simple meals create a feeling of unity because people from different backgrounds eat together. However, large celebrations can create plastic waste and noise pollution. We can reduce these problems through careful planning. Organisers can provide drinking-water stations and encourage people to carry reusable bottles. Food can be served on washable plates or compostable leaf plates instead of single-use plastic. Separate bins should be placed along the route for recyclable and organic waste. Volunteers can guide visitors and keep emergency pathways clear. Loudspeakers should be used at a reasonable volume and only during permitted hours. The procession route can be cleaned immediately after the event, and food waste can be sent for composting. Local shops can be encouraged to avoid plastic bags during the festival. These steps do not reduce the devotion or excitement of the celebration. Instead, they show respect for the neighbourhood and the environment. A temple car festival becomes more meaningful when people celebrate together while keeping their streets clean, safe and peaceful.

Question 3: Regional Folk Art and Handicraft Fairs

A regional folk art and handicraft fair is a colourful place where people can experience the culture of a community. The fair usually includes traditional music, folk dances, handmade products, pottery, weaving and woodwork. Local artists display the skills that have been passed down through generations. Visitors can watch artisans at work and understand the time and effort behind each product. Food stalls add to the atmosphere by serving regional delicacies such as millet dishes, traditional snacks and local sweets. These fairs also give small businesses and craftspeople an opportunity to earn an income. Modern technology can make such events more sustainable. Solar micro-grids can provide electricity for lighting, small stalls and information counters. Energy-efficient lights can reduce power consumption, while rechargeable equipment can be used wherever possible. Organisers can place clearly labelled bins for paper, metal, plastic and food waste. Leftover food and suitable organic waste can be collected for composting. Vendors can use reusable plates, cloth bags and paper packaging instead of disposable plastic. Visitors can be encouraged to use public transport or shared vehicles to reduce traffic. Digital tickets and online maps can also reduce unnecessary paper use. Most importantly, local artisans should be involved in planning the event so that modern methods do not weaken traditional practices. A well-managed fair can protect cultural heritage and the environment at the same time. It allows people to enjoy art, food and community life while supporting a cleaner future.

TOPIC 3 – FREE WRITING: FOOD AND TECHNICAL TRAVEL PLANS

Question 1: Technical Study Tour to a Hydroelectric Dam

Our department can organise a one-day technical study tour to a major hydroelectric dam to understand how water is converted into electrical energy. We can begin early in the morning by travelling in a college bus. Before departure, the faculty coordinator should confirm the visitor permission, attendance list and emergency contact details. We should carry college identity cards, drinking water, simple food and notebooks. After reaching the dam, the first stop can be the visitor information centre, where an engineer explains the purpose of the dam and the basic working of a hydroelectric plant. Next, we can observe the reservoir, intake structure and penstock from the approved visitor areas. The guide can explain how flowing water turns the turbine and how the generator produces electricity. If access is permitted, we can also learn about the control room, transformers and power transmission system through a

demonstration. We should never enter restricted areas or touch equipment. During the journey, we can enjoy local food such as idli, dosa, lemon rice or a regional meal at a clean restaurant. We must keep the surroundings clean and avoid wasting food. Students should follow all safety instructions, stay with their group and wear the protective equipment provided by the facility. At the end of the visit, each student can write a short report about energy conversion, water management and environmental protection. The tour will connect classroom concepts with real industrial practice and help us understand the importance of renewable energy.

Question 2: Study Tour to a Solar and Wind Hybrid Energy Farm

A study tour to a solar and wind hybrid energy farm would help engineering students understand two important renewable energy sources. The group can travel by college bus after confirming permission with the facility. The faculty coordinator should prepare the student list, transport plan, food arrangements and emergency contacts in advance. We can start early so that the group reaches the site before the afternoon heat. On arrival, a site engineer can introduce the layout and explain how solar panels convert sunlight into electricity. Students can observe the photovoltaic arrays from the permitted pathways and learn about panel orientation, inverters and monitoring systems. The next stop can be the wind turbine viewing area. The guide can explain how wind turns the blades and how the generator produces electrical power. We can also learn how both sources are connected to the grid and how energy output changes with weather conditions. During the journey, we can try local dishes such as idli, dosa, millet meals or regional snacks at a hygienic restaurant. We should carry reusable water bottles and avoid littering. For safety, students must stay behind barriers, follow the guide's instructions and never climb or touch the equipment. Hats, comfortable footwear and drinking water are useful during the outdoor visit. Before returning, we can discuss the advantages and limitations of combining solar and wind power. The visit will make renewable energy concepts easier to understand and show how clean electricity can support a sustainable future.

Question 3: Technical Visit to a Seawater RO Desalination Plant

A technical visit to a seawater reverse osmosis plant can help students understand how seawater is treated to produce usable water. We can travel by college bus to the coastal facility after obtaining permission and confirming the visitor schedule. The faculty coordinator should arrange identity cards, attendance, transport and safety instructions. At the plant, an engineer can first explain the need for desalination in water-scarce areas. We can then observe the seawater intake pumping station from the designated visitor area. The guide can describe how screens remove larger materials before the water enters the treatment process. At the next section, students can learn about pre-treatment filters, high-pressure pumps and reverse osmosis membrane racks. The engineer can explain how pressure helps separate fresh water from dissolved salts. We can also learn how the treated water is checked for quality before distribution and how the remaining concentrated brine is managed. During the trip, we can enjoy coastal food such as idli, fish curry or a vegetarian regional meal at a clean restaurant. Students should follow all site rules, wear the required protective equipment and stay away from restricted machinery and wet operating areas. We must not touch pipes, valves or control panels. We should also avoid littering near the coast. After the visit, students can prepare a report on water treatment, energy use and environmental concerns. This tour will show how engineering helps address an important everyday need.

TOPIC 4 – FORMAL LETTERS OF REQUEST

Question 1: Permission for After-Hours Lab Access – AI Hackathon

From:
Harish M.
B.E. CSE – C
St. Joseph’s College of Engineering

To:
The Head of the Department
Department of Computer Science and Engineering
St. Joseph’s College of Engineering

Subject: Request for after-hours laboratory access for National AI Hackathon preparation

Respected Sir/Madam,

I am writing on behalf of my project team to request permission to use the computer laboratory after regular college hours to prepare for the National AI Hackathon. Our team plans to develop an AI-based project that can solve a practical problem through data analysis and intelligent automation. We need additional time to discuss the design, write and test the code, and prepare our final demonstration.

We request access to the laboratory on selected days, at timings approved by the department. We would also be grateful if the required software tools, such as Python, suitable machine-learning libraries, a code editor and internet access, could be made available. We assure you that we will follow laboratory rules, maintain discipline, protect the equipment and leave the room clean after every session. We are also willing to work under the supervision of a faculty member.

Kindly consider our request and grant permission if possible. Your support will help us prepare well and represent our college effectively.

Thank you.

Yours faithfully,
Harish M.
On behalf of the project team

Question 2: Request for an Industrial Visit for 50 Students

From:
Harish M., Student Representative
B.E. CSE – C
St. Joseph's College of Engineering

To:
The Plant Manager
Industrial Robotics and Automation Factory

Subject: Request for permission to visit your industrial facility

Respected Sir/Madam,

On behalf of the first-year engineering students of our college, I kindly request permission to arrange an industrial visit to your robotics and automation factory for a group of 50 students, accompanied by faculty members.

The purpose of this visit is to help students understand how industrial robots are used in manufacturing and how automated systems improve productivity, accuracy and safety. We are especially interested in learning about robotic arms, sensors, programmable controllers, conveyor systems and quality inspection methods. The visit would give us a useful opportunity to connect the concepts taught in our classes with real industrial applications.

We are willing to follow the date, time and visitor rules suggested by your organisation. We will ensure that all students remain with the group, wear the required safety equipment and respect restricted areas. We would appreciate a short technical briefing and a guided tour, if available.

Kindly consider our request and let us know whether the visit can be arranged. Thank you for your time and support.

Yours faithfully,
Harish M.
Student Representative

Question 3: Request for Machine Shop Access – Electric Go-Kart

From:
Harish M.
B.E. CSE – C
St. Joseph's College of Engineering

To:
The Head of the Department
Department of Mechanical Engineering
St. Joseph's College of Engineering

Subject: Request for after-hours machine shop access for an electric go-kart project

Respected Sir/Madam,

I am writing on behalf of our student team to request permission to use the machine shop after regular college hours. We are preparing an energy-efficient electric go-kart for an inter-collegiate design challenge. We need access to suitable workshop facilities to fabricate selected parts, assemble components and carry out basic testing under supervision.

The project will help us learn about design, material selection, manufacturing processes, electrical integration and teamwork. We request permission to use the approved tools and machines at timings convenient to the department. We also request guidance from a faculty member or workshop technician whenever machine operation is required.

We assure you that all team members will follow workshop safety rules, wear the required protective equipment and use machines only with permission. We will maintain the equipment properly and clean the work area after each session. We will also follow the department's instructions regarding materials and storage.

Kindly consider our request and grant us permission to use the facilities. Your support would be very helpful in completing the project successfully.

Thank you.

Yours faithfully,
Harish M.
On behalf of the project team

Question 4: Request for Sponsorship and Duty Leave

From:
Harish M.
Student Representative, Multidisciplinary Team
St. Joseph's College of Engineering

To:
The Principal / Dean of Student Affairs
St. Joseph's College of Engineering

Subject: Request for financial sponsorship and duty leave to attend a national innovation expo

Respected Sir/Madam,

We are pleased to inform you that our five-member multidisciplinary student team has been selected to present a drone prototype at a National Defense Innovation Expo. This is a valuable opportunity for us to demonstrate our technical work, receive expert feedback and represent our college at a national-level event.

We kindly request financial support to help meet the registration, travel and accommodation expenses. We also

request duty leave for the team members for the official dates of the event, so that we can participate without being marked absent. We are ready to submit the selection confirmation, event schedule and an estimated budget for your review.

Our team will represent the institution responsibly and follow all event rules. We will also share a brief report and presentation about our experience after returning. We believe that this opportunity will encourage other students to take part in technical activities and innovation programmes.

We would be grateful if you could consider our request and provide the necessary approval and support.

Thank you for your consideration.

Yours faithfully,
Harish M.
On behalf of the five-member team

TOPIC 5 – COMPARE AND CONTRAST ESSAYS

Question 1: Electric Vehicles and Conventional ICE Vehicles

Electric vehicles (EVs) and conventional internal combustion engine (ICE) vehicles are both used for daily transportation, but they work in different ways. EVs use electric motors and rechargeable batteries, whereas ICE vehicles burn petrol or diesel to produce power. Each type has its own advantages and limitations.

One major advantage of EVs is that they produce no tailpipe emissions while driving. This can help improve air quality in cities. In contrast, petrol and diesel vehicles release exhaust gases during operation. EVs also have fewer moving parts in their drive systems, so routine maintenance can be simpler and sometimes less expensive. On the other hand, the battery is costly and its capacity may reduce over time. Battery production also requires raw materials and energy, which creates an environmental footprint.

ICE vehicles are often easier to refuel because petrol stations are widely available. They can also be convenient for long journeys in places where charging stations are limited. However, fuel costs and engine maintenance can increase their long-term running expenses. EVs may have lower running costs, especially when charging is affordable, but charging time and network availability remain important concerns. Fast chargers are not equally available in every region.

The initial purchase price of an EV may be higher than that of a similar ICE vehicle, although prices and incentives vary. Overall, EVs are a promising option for cleaner transport, while ICE vehicles still offer convenience in areas with limited charging facilities. The better choice depends on budget, travel needs, electricity access and local infrastructure.

Question 2: Remote Internships and On-Site Internships

Remote and on-site internships both help engineering students gain practical experience, but the learning environment is different. A remote internship allows students to work from home or another location using online tools. An on-site internship requires students to visit the workplace and interact with the team in person.

The biggest advantage of a remote internship is flexibility. Students may save travel time and can sometimes manage their work around classes more easily. It also helps them improve their communication and collaboration skills through digital platforms. However, remote interns may find it difficult to access laboratory equipment, machines or physical prototypes. They may also miss informal conversations that help new employees understand workplace culture.

On-site internships provide better opportunities for hands-on learning, especially in fields that involve testing, manufacturing and hardware. Students can observe real processes, ask questions directly and learn by working beside experienced engineers. Face-to-face interaction can also make mentorship and professional networking easier. The disadvantages are travel expenses, fixed working hours and the time needed to reach the workplace. Some students may find it difficult to balance these demands with their academic schedule.

Both formats can be useful when the tasks are planned properly. Remote internships are suitable for coding, documentation, data analysis and other digital work. On-site internships are especially valuable when practical equipment and direct observation are important. A hybrid internship can combine the strengths of both methods. For early-stage engineering students, the best choice is the one that provides meaningful guidance, suitable tasks and a chance to apply classroom knowledge.

Question 3: Cloud Computing and On-Premise Data Centres

Cloud computing and on-premise data centres are two common ways for organisations to store data and run applications. Cloud services use computing resources provided over the internet, whereas on-premise systems are installed and managed at the organisation's own location. Both approaches have benefits and drawbacks.

Cloud computing usually requires less initial investment because a company does not need to purchase and install all the physical servers. It can rent storage and computing power as needed. This makes it easier to increase or reduce capacity when workloads change. In contrast, an on-premise data centre requires a larger upfront investment in servers, networking equipment, cooling and physical space. However, the organisation has direct control over its hardware and configuration.

Security and compliance are important in both models. Cloud providers offer security tools and specialist support, but organisations must still configure access controls and protect their data. Some businesses may have strict rules about where information is stored. An on-premise system can provide greater physical control, although the organisation is responsible for maintenance, updates and security protection.

Cloud services have recurring subscription or usage charges. These costs can become high if resources are not monitored carefully. On-premise systems also have continuing expenses, including electricity, cooling, repairs and staff. In conclusion, cloud computing is useful for flexible workloads and quick expansion, while on-premise systems suit organisations that need direct control over their infrastructure. Many businesses use a hybrid model to balance flexibility, cost and compliance.

Question 4: 3D Printing and Traditional CNC Milling

3D printing and CNC milling are two important methods used to make engineering prototypes. 3D printing, also called additive manufacturing, builds an object layer by layer. CNC milling is a subtractive process in which a cutting tool removes material from a solid block. Both methods are useful, but they suit different types of work.

A major advantage of 3D printing is that it can create complex shapes, hollow structures and internal channels that may be difficult to produce through ordinary machining. It also uses material mainly where it is needed, so it can reduce waste for certain designs. CNC milling, however, often produces more accurate dimensions and a better surface finish, especially when the machine and cutting tools are properly selected. It is suitable for parts that need close tolerances.

For a single prototype or a small number of complex parts, 3D printing can be convenient because it does not always require special cutting fixtures. However, printing can take a long time, depending on the size and layer resolution. CNC milling may be faster for simple parts and can be more efficient for some batch production tasks. On the other hand, milling removes material and may create more waste, especially when the part has a complex shape.

The choice depends on the design, material, quantity, accuracy and available equipment. 3D printing is a strong option for complex shapes and quick design changes. CNC milling is often preferred when surface quality, strength and dimensional accuracy are important. Engineers may even combine both methods during product development.

TOPIC 6 – PERSONAL EMAIL WRITING

Question 1: Email to a High School Mentor

To: mentor@example.com
Subject: My First Semester at Engineering College

Dear Sir,

I hope you are doing well. I wanted to share how my first semester at engineering college has been so far. The experience is quite different from school. I have more independence now, and I am learning to manage my time, attend classes regularly and complete my work without being reminded every day.

The laboratory sessions have been one of the most interesting parts of college. We have started learning basic programming and working on small practical tasks. At first, some of the concepts felt difficult, but practising them regularly has made me more confident. I am also getting used to working with classmates and discussing solutions as a team.

My end-semester examinations are coming up, so I have started revising each subject. I am preparing short notes, practising programming questions and trying to clear my doubts before the exams. I know I still have a lot to learn, but I am enjoying this new stage of my education.

Thank you for the guidance you gave me in school. Your advice about being consistent has been useful in college too. I hope to visit you during the holidays and tell you more about my experience.

Yours sincerely,
Harish

Question 2: Email to a High School Science Teacher

To: science.teacher@example.com
Subject: Updates from My First Term in Engineering

Dear Ma'am,

I hope you are doing well. I am writing to tell you about my first term as an engineering student. College has been a new experience, especially because the subjects are connected to practical work as well as theory. The laboratory classes have helped me understand why concepts are important and how they are used in real situations.

I have also started exploring technical activities outside the classroom. Taking part in a technical club activity has given me a chance to learn from other students and work as part of a team. I am still a beginner, but these experiences have encouraged me to improve my skills and try new things.

As the university examinations are approaching, I have made a simple study plan. I revise the topics taught each day, practise important questions and spend extra time on subjects that I find difficult. I am also completing my lab records and preparing for practical examinations so that I do not have to rush at the last minute.

I often remember the way you encouraged us to ask questions and understand concepts instead of simply memorising them. That advice is helping me now. Thank you for being a supportive teacher.

With regards,
Harish

Question 3: Email to an Elder Sibling or College Alumnus

To: sibling@example.com
Subject: Need Your Advice on Managing College Work

Hi,

I hope everything is going well. College has been interesting, but I am finding it a little challenging to balance classes, assignments and technical activities. My day is usually filled with lectures and laboratory sessions. After college, I try to

revise the topics covered in class and complete the work given by the teachers.

One difficulty is managing lab record submissions along with regular assignments. Sometimes several deadlines come close together, and I find it hard to decide what to complete first. I also want to take part in technical clubs because I know they can help me improve my practical skills. At the same time, I do not want these activities to affect my studies.

Since you have already experienced college life, I wanted to ask for your advice. How did you plan your daily schedule? Did you set aside a fixed time for lab records and revision? I would also like to know how you prepared for semester examinations without feeling too stressed at the last moment.

I am trying to build a routine that I can follow consistently. Your suggestions would really help me manage my time better and make good use of college opportunities.

Thanks in advance. Talk to you soon!

With love,
Harish

TOPIC 7 – TRAVEL PLANNING AND BUDGETARY ITINERARIES

Question 1: Four-Day Tour to an Aerospace and Satellite Launch Station

A four-day technical tour to an aerospace and satellite launch station would give engineering students a useful introduction to space technology. The college can arrange a hired bus or train travel, depending on the distance. Bookings should be confirmed early, and students should carry identity cards, permission letters and emergency contact details.

Day 1: Travel to the destination, check into the accommodation and attend an introductory briefing about the launch station and its safety rules.
Day 2: Visit the visitor gallery and learn about launch vehicles, satellite structures and the stages of a launch. Attend a presentation by technical staff.
Day 3: Observe approved display areas related to satellite testing, tracking and mission control. Students should not enter restricted zones or photograph areas where photography is prohibited.
Day 4: Attend a final discussion, prepare notes and return to college.

An estimated budget per student may be: transport ■2,000, accommodation ■2,400, food ■1,200, local travel ■400 and entry or activity charges ■500. The total estimated cost is ■6,500. The actual amount may change depending on distance, group size and booking rates. Students must follow the instructions of staff, remain with their group and wear any required protective equipment. They should drink enough water, keep the premises clean and avoid touching equipment. This tour can help students connect classroom science with real aerospace applications.

Question 2: Four-Day Tour to a Commercial Port and Maritime Dry Dock

A four-day industrial exposure tour to a commercial port and maritime dry-dock facility can help engineering students understand shipping, logistics and ship maintenance. The college can arrange train or bus travel and book accommodation close to the port. All visitor permissions should be obtained in advance.

Day 1: Travel to the port city, check into the accommodation and attend a briefing about port operations and visitor safety.
Day 2: Visit the approved container terminal viewing area. Learn how containers are identified, moved and recorded. Observe the role of cranes and transport vehicles from a safe location.
Day 3: Visit the dry-dock facility with a guide. Learn about ship inspection, repair planning, corrosion protection and maintenance procedures. Students must remain in authorised walkways.
Day 4: Attend a question-and-answer session, complete the visit report and return.

The estimated cost per student is ■1,800 for travel, ■2,400 for accommodation, ■1,400 for food, ■500 for local transport and ■400 for entry or other charges. The total is approximately ■6,500. These figures are only planning

estimates and should be confirmed before booking. Students should wear safety shoes, helmets or reflective jackets when instructed. They must stay away from moving vehicles, lifting operations and restricted dock areas. No one should touch machinery or enter the water-side edge. The tour will provide a practical understanding of port engineering and maritime operations.

Question 3: Four-Day Tour to an Electric Locomotive Works and Railway Signalling Centre

A four-day technical tour to an electric locomotive manufacturing works and an automated railway signalling centre can help students understand modern railway technology. The group can travel by train, with tickets booked in advance. The faculty coordinator should arrange accommodation, local transport and official visitor permission.

Day 1: Travel to the destination, check into the accommodation and attend a briefing about the tour schedule and safety rules.
Day 2: Visit the approved locomotive manufacturing areas. Learn about the traction motor, bogie, braking system and electrical control equipment through demonstrations.
Day 3: Visit the railway signalling centre. Engineers can explain how signals, track circuits, control systems and communication networks help trains move safely. Students should observe only from authorised areas.
Day 4: Review the technical observations, complete the visit report and return to college.

An estimated per-student budget is ■1,500 for train travel, ■2,400 for accommodation, ■1,400 for food, ■500 for local transport and ■400 for other charges. The estimated total is ■6,200. Actual expenses may vary with the route and booking dates. Students must carry their identity cards, follow staff instructions and wear the required protective equipment. They must never cross railway tracks, touch electrical equipment or enter restricted work areas. The tour will help students understand how mechanical, electrical and computer systems work together in modern rail transport.

PROJECT PROPOSAL

SMART AMBULANCE ROUTE SYSTEM

An integrated approach to emergency ambulance routing and hospital selection

Project Domain	Healthcare Technology / Intelligent Transportation
Project Type	Prototype and Ideation
Department	Computer Science and Engineering

Prepared as a project ideation document

1. Abstract

The Smart Ambulance Route System is a proposed digital solution designed to support faster and more informed decisions during medical emergencies. In an emergency, reaching a hospital quickly is important, but choosing a hospital that can provide the required treatment is equally important. Traffic congestion, delays in communication and a lack of updated information about ambulance and hospital availability can make this process difficult.

The proposed system brings together information about an emergency location, available ambulances, road traffic and nearby hospitals. Based on the available information, it recommends a suitable route and a hospital that is both reachable and capable of handling the emergency. The project will be developed as a prototype to demonstrate the complete flow from an emergency request to ambulance routing and hospital selection.

2. Introduction

Ambulances play a critical role in emergency healthcare. However, an ambulance may lose valuable time when the route is congested or when the destination hospital is not prepared to handle the patient's condition. In many situations, information about traffic, ambulance availability and hospital facilities is managed separately. This can make coordination slower and less effective.

The Smart Ambulance Route System aims to connect these important pieces of information through one coordinated platform. Instead of considering distance alone, the system considers route conditions and the availability of emergency services at hospitals. The goal is to support emergency response teams with useful recommendations, while keeping the process simple and understandable.

3. Problem Statement

- Traffic congestion can delay ambulances, especially during busy hours.
- Manual communication between emergency callers, ambulance services and hospitals may take additional time.
- The nearest hospital may not have the required emergency department, equipment or available capacity.
- Ambulance location, traffic conditions and hospital readiness may not be available in one place.
- Choosing a destination based only on distance may lead to an unsuitable hospital selection.

4. Objectives

- To create a prototype that connects emergency requests, ambulances, routes and hospitals.
- To identify an available ambulance for an emergency request.
- To consider traffic conditions while recommending a route.
- To help select a hospital based on both distance and emergency-service availability.
- To display the recommended route, destination hospital and current response status in a clear way.

5. Proposed Solution

The proposed platform will accept an emergency request with the patient's location and basic emergency details. It will then use the available ambulance information to identify a suitable ambulance. The system will compare possible routes using traffic information and evaluate nearby hospitals based on their distance and emergency-service availability. Finally, it will display a recommended route and destination hospital, along with the status of the ambulance response.

The system is intended to assist decision-making. It does not replace emergency medical professionals, ambulance dispatchers or hospital staff. The quality of its recommendations will depend on the accuracy and freshness of the information supplied to the prototype.

6. Main Modules

Module	Purpose
Emergency Request Module	Collects the emergency location and basic information about the situation.
Ambulance Availability Module	Displays available ambulances and helps identify one that can respond to the request.
Traffic and Route Module	Uses route and traffic information to compare possible paths and recommend a suitable route.
Hospital Availability Module	Stores or displays hospital details, including distance and whether the required emergency service is available.
Recommendation Module	Combines the available information to suggest a route and a suitable destination hospital.
Status Tracking Module	Displays the progress of the emergency request and ambulance response.

7. Working Process

- **Step 1:** An emergency request is entered with the incident location and basic emergency details.
- **Step 2:** The system checks the ambulance information and identifies an available ambulance.
- **Step 3:** Possible routes are reviewed using the traffic information available to the prototype.
- **Step 4:** Nearby hospitals are checked for distance and the availability of the required emergency service.
- **Step 5:** The system recommends a suitable route and destination hospital.
- **Step 6:** The ambulance response status is displayed so that the progress can be followed.

8. Example Scenario

Assume an emergency request is received and three hospitals are available in the surrounding area. The sample information below shows why distance alone should not decide the destination.

Hospital	Distance	Emergency availability	System consideration
Hospital A	3.2 km	Available	A possible suitable destination
Hospital B	2.4 km	Not available	Not selected despite being closer
Hospital C	5.1 km	Available	An alternative if route or capacity makes it preferable

In this example, Hospital B is the closest but does not have emergency availability. Hospital A may therefore be a more appropriate option, subject to route conditions and the actual needs of the patient. The prototype demonstrates how hospital readiness can be considered along with distance.

9. Detailed System Design

This section expands the working process into a design that can be built and tested as a prototype. It describes the information the system stores, how it selects an ambulance and a hospital, and how it behaves when data is missing. All values shown are sample values for demonstration.

9.1 Core Data Entities

Entity	Key information held
Emergency Request	Request ID, incident location, emergency category, patient notes, time received, status.
Ambulance	Ambulance ID, type (basic or advanced), current location, availability status, crew contact.
Hospital	Hospital ID, name, location, emergency services offered, capacity indicator, time of last update.
Route	Start and end points, distance, estimated travel time, traffic level along the path.
Status Record	Request ID, assigned ambulance, selected hospital, response stage and the time of each stage.

9.2 Ambulance Selection

The system first filters the ambulance list to units marked as available. For each of these it estimates the travel time to the incident using the route and traffic information. The ambulance with the shortest estimated arrival time is recommended. If two ambulances are close in arrival time, the one whose type best suits the emergency category is preferred.

9.3 Hospital Selection and Scoring

Hospital selection works in two stages. In the first stage, any hospital that does not offer the required emergency service is removed from consideration, however close it is. In the second stage, the remaining hospitals are scored from 0 to 1 on three factors, where a higher score is better:

- **Travel-time score (weight 0.6):** the shortest estimated travel time among the candidates divided by that hospital's own estimated travel time, so traffic is reflected and not just distance.
- **Capacity score (weight 0.3):** the reported availability of beds or emergency capacity at the hospital.
- **Data freshness score (weight 0.1):** how recently the hospital's information was updated; older information scores lower.

The weights above are illustrative starting values. They are meant to be adjusted during testing and reviewed with domain experts before any real use.

Applying this to the scenario in Section 8, with sample travel times and capacity values:

Hospital	Service available	Est. time	Time score	Capacity	Freshness	Final score	Outcome
Hospital A	Yes	9 min	1.00	0.60	1.00	**0.88**	**Recommended**
Hospital B	No	7 min	–	–	–	–	Excluded
Hospital C	Yes	11 min	0.82	0.90	1.00	0.86	Alternative

Hospital B is excluded despite being closest. Hospital A scores slightly higher than Hospital C because its shorter travel time outweighs C's better capacity. The result is shown to the dispatcher with the reason, so the recommendation can be checked and overridden.

9.4 Handling Missing or Outdated Data

- **No ambulance available:** the request is flagged to the dispatcher instead of being silently left waiting.
- **No traffic data:** the route is based on distance and a standard speed estimate, and is marked as lower confidence.
- **Outdated hospital data:** the hospital is still shown, but with a warning that its availability may have changed.
- **No hospital offers the required service:** the system lists the nearest alternatives and asks the dispatcher to decide.

Each of these cases, together with the Section 8 scenario and a congested-route case in which the fastest path changes, will be used as a test scenario to check that the prototype behaves as described.

10. Proposed Technology and Resources

The final technology stack can be selected according to the prototype requirements and the tools available to the project. A possible implementation may include:

- **Frontend:** A simple web or mobile interface for entering emergency details and viewing recommendations.
- **Backend:** Application logic to process requests, ambulance details, route information and hospital data.
- **Database:** A structured store for ambulance records, hospital details, emergency-service availability and request status.
- **Maps and traffic data:** A mapping service or simulated route data for displaying routes and traffic conditions.
- **Testing data:** Sample emergency locations, ambulance records and hospital availability for prototype demonstrations.

11. Expected Outcomes

- A working prototype that demonstrates the emergency request-to-hospital recommendation process.
- A clear view of available ambulance information and response status.
- Route recommendations that take traffic information into account when such data is available.
- Hospital suggestions that consider emergency-service availability rather than distance alone.
- A demonstration of how connected information can support better coordination during emergencies.

12. Scope and Limitations

The initial scope is a prototype that demonstrates the main workflow using sample or available data. Real-time ambulance tracking, live traffic integration and direct hospital-system connectivity may require external services, permissions and reliable data-sharing arrangements. If live integrations are not available, these functions can be represented with simulated data during the demonstration.

The system can only provide useful recommendations when ambulance and hospital information is accurate and up to date. It should be treated as a decision-support tool and not as a replacement for professional medical judgement or official emergency dispatch procedures.

13. Future Enhancements

- Integrate verified real-time traffic and map services.
- Enable GPS-based ambulance location updates.
- Connect with hospitals to receive updated emergency-service and capacity information.
- Add priority handling for different emergency categories.
- Provide notifications to relevant emergency response personnel.
- Evaluate recommendation quality using realistic test scenarios and feedback from domain experts.

14. Conclusion

The Smart Ambulance Route System proposes a coordinated way to support emergency ambulance routing and hospital selection. By bringing together emergency location, ambulance availability, traffic conditions and hospital readiness, the system aims to reduce avoidable delays and help users make more informed decisions. The prototype will focus on demonstrating the complete workflow and showing why the most suitable hospital is not always the nearest one. With reliable data and careful development, the concept could be expanded into a more connected emergency response support platform.

— End of Project Proposal —

LINKEDIN PROFILE

HARISH.M

B.E. CSE Student | Aspiring Software Developer | AI & Machine Learning Enthusiast | Python • C • SQL

1. Header & Intro Section

Name: Harish.M
Headline: B.E. CSE Student | Aspiring Software Developer | AI & Machine Learning Enthusiast | Python • C • SQL
Education: B.E. Computer Science and Engineering – St.Joseph's college of Engineering
Location: Chennai, Tamil Nadu, India
LinkedIn: https://www.linkedin.com/in/harish-manivannan-43176843b/
GitHub: https://github.com/Harish1717git

2. About Section

I am a first-year Computer Science and Engineering student with a strong interest in software development, programming, and emerging technologies. I am building my foundation in Python, C, and SQL while improving my problem-solving and logical thinking skills through academic and personal projects.

I enjoy using technology to solve real-world problems and create software solutions that can make people's lives easier. I have also explored project-based learning through initiatives such as the Smart Ambulance Route System, which focuses on using technology to support better decision-making during emergencies.

I am interested in Software Development and Artificial Intelligence & Machine Learning and enjoy learning new technologies, working on innovative projects, and continuously improving my technical skills.

My goal is to keep learning, gain practical experience, build meaningful projects, and grow into a skilled software professional who can create useful solutions to real-world problems.

3. Featured Section

GitHub Profile

My GitHub profile showcasing programming practice, projects, and my learning journey.

Smart Ambulance Route System

A project demonstrating a real-world problem-solving approach using technology to support faster and better emergency decisions.

4. Education

Bachelor of Engineering – Computer Science and Engineering

St.Joseph's college of Engineering
2026–2030 (Expected)

Currently pursuing a B.E. in Computer Science and Engineering, building a strong foundation in programming, problem-solving, software development, and emerging technologies.

5. Projects / Experience

Smart Ambulance Route System — Project / Hackathon

- Developed a decision-support concept for emergency ambulance routing.
- Connects emergency requests with ambulance availability, traffic conditions, route information, and hospital availability.
- Recommends a suitable route and hospital by considering both distance and emergency-service availability.
- Explores how software solutions can support faster and better emergency decisions.
- Provides practical exposure to real-world problem-solving through project-based learning.

6. Courses & Certifications

SWAYAM – Computing Fundamentals

Currently Pursuing | 2026

Focus: Computing fundamentals, programming concepts, and foundational computer science knowledge.

No additional certifications are included because only completed or currently pursuing learning activities are listed.

7. Core Skills

Technical Skills

- Python
- C
- SQL
- Problem Solving

Professional Skills

- Communication
- Teamwork
- Adaptability
- Continuous Learning

Areas of Interest

- Software Development
- Artificial Intelligence & Machine Learning
- Programming
- Real-World Problem Solving

8. Achievements & Activities

- Participated in project-based learning through the Smart Ambulance Route System.
- Explored a real-world emergency transportation problem through a software-oriented project.
- Developing programming and problem-solving skills through academic and personal projects.
- Continuously learning and exploring Software Development and AI & Machine Learning.

9. Professional Pitch

Good morning everyone. My name is Harish.M, and I am a first-year Computer Science and Engineering student at St.Joseph's college of Engineering. I am interested in software development and Artificial

Intelligence and Machine Learning. I am currently developing my skills in Python, C, and SQL while improving my problem-solving abilities through academic and personal projects. I have also worked on a Smart Ambulance Route System project that focuses on solving a real-world emergency problem using technology. My goal is to continuously learn, gain practical experience, and become a skilled software developer who creates solutions that make people's lives easier. Thank you.

10. Recommendations

To be added as I gain academic, project, and professional experience. No fictional recommendations are included.

11. Honors, Awards & Community Engagement

Project & Academic Activities

- Participated in project-based learning and explored real-world problem-solving through the Smart Ambulance Route System.
- Developing technical, communication, and teamwork skills through academic activities and projects.

12. Contact & Profile Links

Email: harishmanivannan08@gmail.com

Phone: 9789260205

Location: Chennai, Tamil Nadu, India

LinkedIn: https://www.linkedin.com/in/harish-manivannan-43176843b/

GitHub: https://github.com/Harish1717git

GITHUB PROFILE README

Hi there, I'm Harish M

First-Year CSE Student | Aspiring Software Developer | AI & Machine Learning Enthusiast

WHAT I STUDY

Pursuing a **Bachelor of Engineering (B.E.) in Computer Science and Engineering** at **St. Joseph's College of Engineering, Chennai**.

CORE TARGET

Working toward becoming a **Software Developer**, with a long-term interest in **Artificial Intelligence & Machine Learning**.

PROJECT FOCUS

I enjoy exploring how programming and software can be used to solve real-world problems. My current project work includes the **Smart Ambulance Route System**, which focuses on supporting better emergency routing decisions.

CURRENTLY LEARNING

Python, C, SQL, problem-solving, software development fundamentals, Git/GitHub, and the foundations of AI & Machine Learning.

ABOUT ME

I am a first-year Computer Science and Engineering student building my technical foundation through programming, academic learning, and practical projects.

I am currently developing my skills in **Python, C, and SQL** while improving my logical thinking and problem-solving abilities. I enjoy learning by experimenting with code and applying what I learn to practical ideas.

My main areas of interest are **Software Development, Artificial Intelligence, Machine Learning, and real-world problem solving**.

This GitHub profile is a record of my learning journey from beginner programming practice to projects that I build as my skills grow.

ASK ME ABOUT

- Python fundamentals
- C programming
- SQL basics
- Problem solving
- Beginner software projects
- Git & GitHub
- My interest in AI & ML

FUN FACT

I enjoy taking a real-world problem and thinking about how software could be used to make the solution faster, smarter, or more useful.

Harish M — GitHub Profile Portfolio

DOMAIN TECHNOLOGIES & TOOLS

Programming Languages

`Python` `C` `SQL`

Tools & Environments

`Git` `GitHub` `VS Code`

Areas of Interest

`Software Development` `Artificial Intelligence` `Machine Learning` `Problem Solving`

FEATURED REPOSITORY: SMART AMBULANCE ROUTE SYSTEM

A software-oriented project designed to support emergency ambulance routing by considering:

- Emergency location
- Ambulance availability
- Route information
- Traffic conditions
- Hospital availability

The concept focuses on recommending a suitable route and hospital by considering **distance together with emergency-service availability**, rather than selecting a hospital based on distance alone.

Repository: smart-ambulance-route-system

LEARNING JOURNEY

Completed

- Started B.E. CSE journey
- Created LinkedIn and GitHub profiles
- Built first public GitHub project
- Started developing fundamentals in Python, C, and SQL

In Progress & Next Goals

- Strengthening C and Python skills
- Learning Git/GitHub workflows
- Exploring AI & ML fundamentals
- Practicing Data Structures & Algorithms

CAREER ROADMAP

Phase / Years	Focus & Key Objectives
2026–2027 Foundation	Build strong fundamentals in programming, problem solving, SQL, Git/GitHub, and core computer science concepts.
2027–2028 Development	Build more practical projects and begin deeper exploration of Artificial Intelligence and Machine Learning.
2028–2029 Practical Experience	Work on larger projects, strengthen software development skills, and seek opportunities for practical industry exposure.

Harish M — GitHub Profile Portfolio

Phase / Years	Focus & Key Objectives
2029–2030 Career Preparation	Graduate with strong technical foundations and prepare for a professional career in Software Development and AI/ML.

EDUCATION

Bachelor of Engineering (B.E.)

Computer Science and Engineering

St. Joseph's College of Engineering, Chennai

2026 – 2030 (Expected)

LEARNING & COURSES

SWAYAM

Computing Fundamentals

Currently pursuing

"Learn • Build • Improve • Grow"

CONNECT WITH ME

Email: harishmanivannan08@gmail.com

LinkedIn: Harish M

GitHub: Harish1717git

INTERCHANGE

During this course, our class used the Cambridge One platform for English communication and lab activities, working through Interchange Fifth Edition Level 2. Instead of relying only on textbooks or classroom explanations, we practised English through online units, exercises, and tests. This gave me a structured, self-paced way to improve my language skills and to see my progress regularly.

My Progress

I completed 141 out of 240 exercises in the Digital Workbook. Of these, 118 were above the target score and 23 were below target, with an average score of 87% and a total of 5 hours 29 minutes spent practising.

Completed	Above Target	Below Target	Average Score	Time Spent
141 / 240	118 / 141	23 / 141	87%	5h 29m

What I Learnt

- **Listening:** I learned to focus on the main idea and pick out specific details from recorded conversations, announcements, and talks. Being able to replay audio helped me catch things I missed the first time, and with practice I became more comfortable with different accents and speaking speeds.
- **Reading:** I improved my reading speed and comprehension. I learned to skim for the overall idea, scan for particular information, and guess the meaning of unfamiliar words from context instead of stopping at every difficult word.
- **Writing:** I learned how to structure emails, paragraphs, and short pieces of writing with a clear beginning, middle, and end. I also became more careful about punctuation, sentence formation, and choosing the right tone for the situation.
- **Grammar:** I revised important topics such as tenses, articles, prepositions, and sentence structure. The exercises made it clear which mistakes I repeat most often, so I could correct them.
- **Vocabulary:** I learned new words, phrases, and collocations that are useful in everyday, academic, and professional settings, and I now use them more naturally in my speaking and writing.
- **Study Habits:** Beyond language skills, I learned to manage my time, complete tasks regularly, and learn independently through online practice.

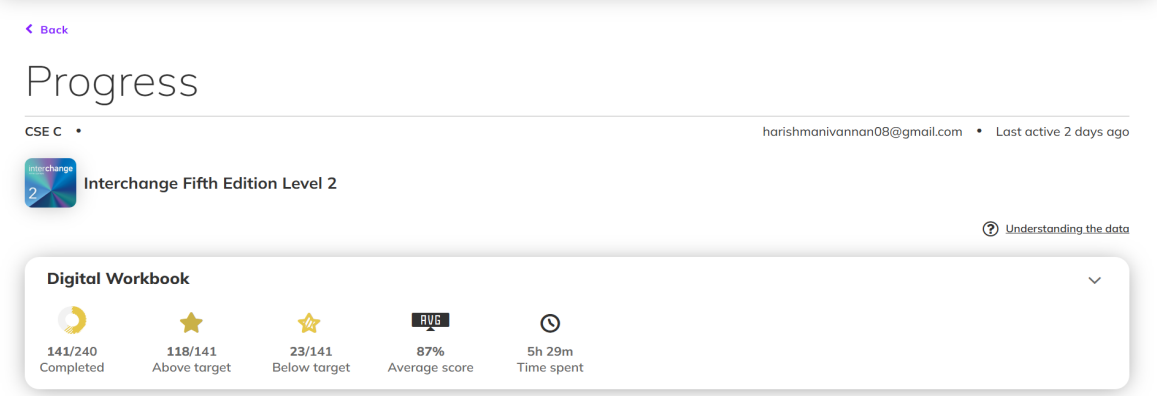
What I Liked About the Platform

- Interactive exercises with instant feedback, so I could see my mistakes and correct them right away.
- The self-paced format, which let me revisit difficult topics as many times as I needed.
- Clear scores and progress tracking that showed where I was improving and where I needed more work.

- Well-organised units that moved from basic to more advanced topics in a logical order.
- Lab sessions that gave me dedicated time to practise, not just theory.

How This Will Help Me in the Future

The skills I developed here will support me in many areas: writing clear emails and reports, understanding spoken and written information in professional settings, preparing for placements and interviews, and completing academic projects. My 87% average score shows that regular practice works, and I plan to keep improving by completing the remaining exercises and continuing to practise English on my own.



THE BOY WHO MEASURED RAIN

Short Story • CEFR Estimate • Coherence • Grammar Analysis

1. Is it a short story or a journal?

Genre: Short story. The writing introduces a main character, describes his curiosity, presents a question, and resolves it through his teacher's explanation. It is not a journal entry because it narrates events involving a character rather than recording personal thoughts in a journal format.

Title note: The title is interesting, but the current plot does not show Mahi measuring rainfall. You could change the title to The Boy Who Wanted to Understand Rain, or add a scene in which he measures rainfall.

2. Estimated CEFR Level

Estimated range: B1–B2 features. The vocabulary and sentence structures show some B2-like features, such as "determined to find the answer," "introduced him to the scientific explanation," and "allowing the cycle to continue." However, this short sample is not enough to confirm an overall B2 level.

Skill	Assessment
Vocabulary	Good variety, with some advanced expressions.
Grammar	Mostly understandable, with a few errors.
Coherence	Good logical sequence.
Cohesion	Good use of transitions.
Natural expression	Generally natural, with a few awkward phrases.

This is an informal estimate, not an official CEFR assessment.

3. Coherence and Cohesion

Coherence: 8/10. The story follows a logical sequence, from the rainfall to Mahi's curiosity and the teacher's explanation.

Cohesion: 7/10. Expressions such as "One afternoon," "Determined to find the answer," and "Eventually" connect the ideas. A stronger ending makes the story feel more complete.

These are informal ratings, not official CEFR scores.

4. Grammatical Errors and Corrections

Original	Correction	Explanation
“THE BOY WHO MEASURED RAIN One fine day”	Put the title on its own line, then start a new paragraph.	Separate the title from the story.
“the Sun heats water”	“the Sun heats the water”	Both can be grammatical; “the water” fits the specific sources already mentioned.
“gather together to form clouds”	“gather to form clouds”	“Gather together” is acceptable but slightly repetitive.
“continue.The boy”	“continue. The boy”	Add a space after the full stop.
“returned hme”	“returned home”	Correct the spelling.
“with gaining knowledge”	“having gained new knowledge”	Use a natural grammatical construction.

Corrected final sentence: “The boy thanked his teacher and returned home, having gained a better understanding of the science behind rain.”

5. Polished B2-Style Short Story

THE BOY WHO MEASURED RAIN

One fine day, dark clouds suddenly covered the sky, and rain began to pour over Anna Nagar, Chennai. Among the people watching the unexpected rainfall was a curious seven-year-old boy named Mahi, who was always eager to discover something new. Unlike other children who simply enjoyed playing in the rain, Mahi wanted to understand the science behind it.

Mahi was particularly fond of watching cartoons. One afternoon, he watched an episode of Ninja Hattori that presented a humorous explanation of rainfall. According to his imagination, ninjas were responsible for pouring water from the sky, creating lightning with torches, and producing thunder by beating drums. Although this explanation seemed amusing, he began to wonder whether there was any truth behind it.

Determined to find the answer, Mahi decided to ask his teacher the following day. As soon as he arrived at school, he approached her and explained what he had seen in the cartoon. His teacher smiled at his curiosity and patiently introduced him to the scientific explanation of rainfall.

She explained that rain was an essential part of the water cycle. When the Sun heats water in rivers, lakes, and oceans, some of it evaporates and rises into the atmosphere as water vapour. As the vapour cools, it condenses into tiny water droplets that gather to form clouds. Eventually, when the droplets become heavy enough, they fall to the ground as rain. This water replenishes rivers and lakes, while some of it seeps into the soil. Later, the Sun heats the water again, allowing the cycle to continue.

Mahi listened carefully and was fascinated by how science could explain something that had seemed mysterious to him. He thanked his teacher for the explanation and returned home, having gained a better understanding of the natural world.

That evening, as the rain continued outside, Mahi looked at the falling droplets with a new sense of wonder. He realised that asking questions and seeking reliable explanations could help him discover the truth behind everyday events. What had begun as a simple question about rain had become an exciting lesson in science.

From that day onwards, Mahi remained curious about the world around him, always eager to explore, observe, and learn something new.

6. Final Evaluation

Category	Evaluation
Genre	Short story: it has a character, a clear sequence of events, and a conclusion.
Theme	Curiosity and scientific learning; it encourages readers to ask questions and learn through observation.
Main improvement	Balance the scientific explanation with Mahi’s feelings and actions so the piece feels like a story rather than only an explanation of the water cycle.

Writing tip: Use this polished version to study the improvements, then practise rewriting the story in your own words. Your CEFR level is best shown by your independent writing across different tasks.